Program

```
#include <stdio.h>
#define MAX 50
int graph[MAX][MAX];
int visited[MAX];
int n;
void DFS(int vertex)
{
  int i;
  // Mark vertex as visited
  visited[vertex] = 1;
  // Display vertex
  printf("%d ", vertex);
  // Visit adjacent vertices
  for (i = 0; i < n; i++)
  {
    if (graph[vertex][i] == 1 && visited[i] == 0)
    {
      DFS(i);
    }
  }
}
int main()
{
  int i, j;
  int start;
  // Read number of vertices
  printf("Enter number of vertices: ");
  scanf("%d", &n);
  // Read adjacency matrix
  printf("Enter the adjacency matrix:\n");
  for (i = 0; i < n; i++)
  {
     for (j = 0; j < n; j++)
    {
      scanf("%d", &graph[i][j]);
    }
  }
```

```
  // Initialize visited array
  for (i = 0; i < n; i++)
  {
    visited[i] = 0;
  }
  // Read starting vertex
  printf("Enter starting vertex: ");
  scanf("%d", &start);
  printf("DFS Traversal: ");
  DFS(start);
  return 0;
}
```

Sample Input

Using the same graph as the BFS example:

Enter number of vertices: 5

Enter the adjacency matrix:

0 1 1 1 0

1 0 1 0 0

1 1 0 0 1

1 0 0 0 0

0 0 1 0 0

Enter starting vertex: 0

Output

DFS Traversal: 0 1 2 4 3